

Indian Institute of Information Technology, Allahabad

Department of Information Technology

AI-Powered Clinical Assistant for Homeopathic Repertorization Using Kent's Repertory

SUPERVISOR

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REPOSITORY

github.com/su4532/kent-repertory-explorer

PROJECT TEAM

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SUBMISSION DATE

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1. Executive Summary

We propose building an **AI-powered clinical assistant** that transforms the homeopathic consultation workflow. The system introduces an empathetic conversational chatbot that conducts preliminary patient interviews, automatically extracts symptoms using NLP, maps them to Kent's Repertory rubrics via semantic search, and generates a structured pre-consultation report for the doctor.

The core value proposition: A homeopathic doctor currently spends **30–40 minutes** understanding a patient's symptoms. Our system reduces this to **5–10 minutes** by providing a polished, structured symptom report *before* the physical consultation.

The system is grounded in an open-access **digitized Kent's Repertory database** (developed by `su4532` in `kent-repertory-explorer`) containing **74,513 rubrics**, **679 remedies**, and **507,179 rubric-remedy mappings**, which our team transforms into a dense semantic vector index for neural clinical search.

2. Problem Statement

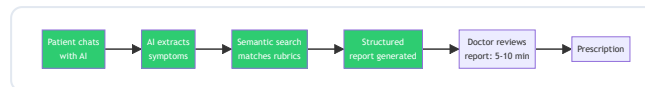
Current Workflow (Without AI)



Pain points:

- 🕒 **Time-intensive:** 30–40 minutes just for symptom gathering, before analysis begins
- 📖 **Information overload:** Kent's Repertory has 74,513 rubrics across 37 sections — manual search is slow and error-prone
- 🧠 **Cognitive burden:** Doctors must hold multiple symptom dimensions (location, sensation, modality, laterality, concomitants) in working memory simultaneously
- ❌ **Inconsistency:** Different practitioners may map the same symptom description to different rubrics

Proposed Workflow (With AI)



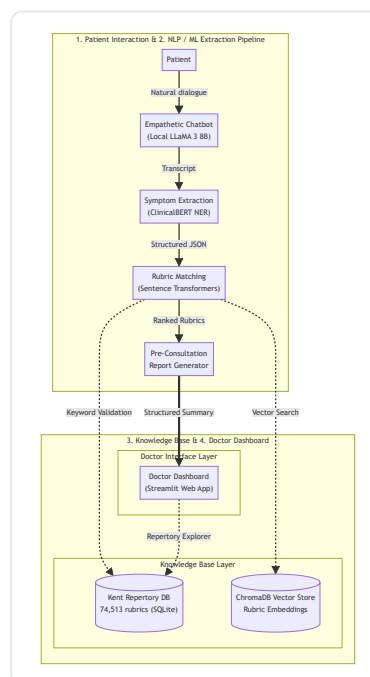
3. System Architecture

3.1 High-Level Architecture



Figure: AI-Powered Homeopathic Clinical Assistant — System Architecture

3.2 Detailed Component Architecture



4. Existing Knowledge Base

Important

Foundation & Attribution: Our project leverages and builds upon the digitized Kent's Repertory open-source foundation developed by [su4532](https://github.com/su4532/kent-repertory-explorer) (github.com/su4532/kent-repertory-explorer). This provides an initial structured SQLite schema with 74,513 rubrics, 679 remedies, and 507,179 mappings. Having this raw digital foundation in place allows our team to focus directly on the **core machine learning challenges**: dense vector representation, ClinicalBERT symptom NER, synthetic case generation, and conversational clinical reasoning.

4.1 Database Statistics

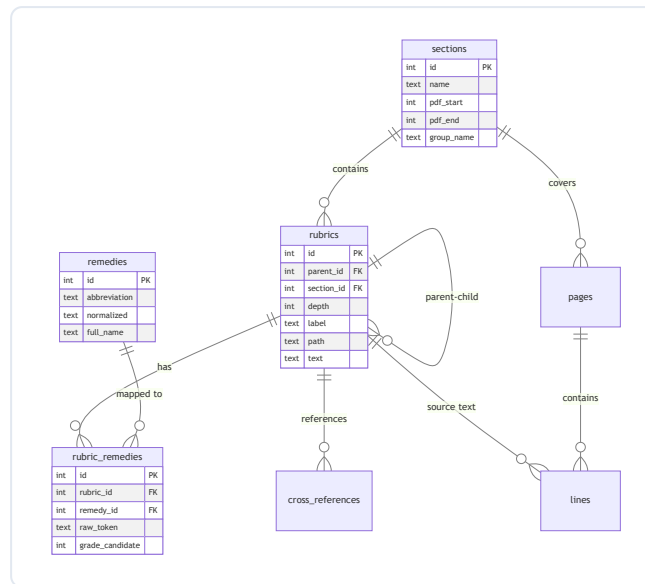
Metric	Value
Total rubrics	74,513
Total remedies	679
Rubric–remedy mappings	507,179
Cross-references	1,923
Sections/chapters	37 (covering all Kent's Repertory content)
Pages digitized	1,469
OCR lines	137,062
Database size	112 MB (SQLite, fully indexed)
Full-text search	FTS5 enabled

4.2 Repertory Coverage — All 37 Sections

#	Section	Rubrics	#	Section	Rubrics
1	MIND	4,933	20	PROSTATE GLAND	93
2	VERTIGO	477	21	URETHRA	599
3	HEAD	7,240	22	URINE	428
4	EYE	1,998	23	GENITALIA-MALE	1,130
5	VISION	954	24	GENITALIA-FEMALE	1,461
6	EAR	2,189	25	LARYNX AND TRACHEA	731
7	HEARING	188	26	RESPIRATION	842
8	NOSE	1,680	27	COUGH	1,555
9	FACE	2,266	28	EXPECTORATION	363
10	MOUTH	1,668	29	CHEST	3,830
11	TEETH	798	30	BACK	4,115
12	THROAT	980	31	EXTREMITIES	17,179
13	THROAT-EXTERNAL	325	32	SLEEP	1,144
14	STOMACH	3,132	33	CHILL	764
15	ABDOMEN	4,095	34	FEVER	583
16	RECTUM	1,415	35	PERSPIRATION	404
17	STOOL	283	36	SKIN	1,301
18	BLADDER	778	37	GENERALITIES	2,336
19	KIDNEYS	256			

Sections 18–22 (Bladder through Urine) are grouped under **Urinary Organs** in the original text. The database correctly represents this grouping via the `group_name` field.

4.3 Database Schema

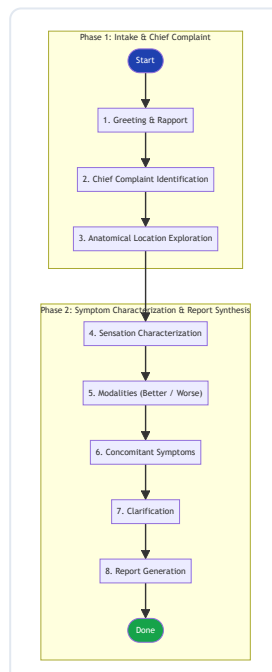


5. Methodology

5.1 Empathetic Chatbot (Conversational Agent)

The chatbot is the patient's first point of contact. Unlike a rigid questionnaire, it conducts a **natural, empathetic conversation** that systematically gathers the symptom dimensions critical to homeopathic case-taking.

Conversation State Machine



Homeopathic Symptom Dimensions Captured

Dimension	What it captures	Example
Location	Body part/region	"Right side of head"
Sensation	Type of feeling	"Pressing, throbbing, burning"
Modality	Aggravation/amelioration	"Worse in morning, better by pressure"
Laterality	Left/right/bilateral	"Left-sided"
Concomitant	Accompanying symptoms	"With nausea and anxiety"
Mental/Emotional	Psychological state	"Irritable, restless at night"
Temporality	Time patterns	"Worse at midnight, periodic"

Architecture: Hybrid LLM + Structured State Tracking

- **LLaMA 3 (8B)** running locally via Ollama handles the natural conversation
- A **structured JSON state tracker** ensures all symptom dimensions are progressively filled
- The system prompt is specifically designed for homeopathic case-taking methodology

5.2 NLP Symptom Extraction Pipeline

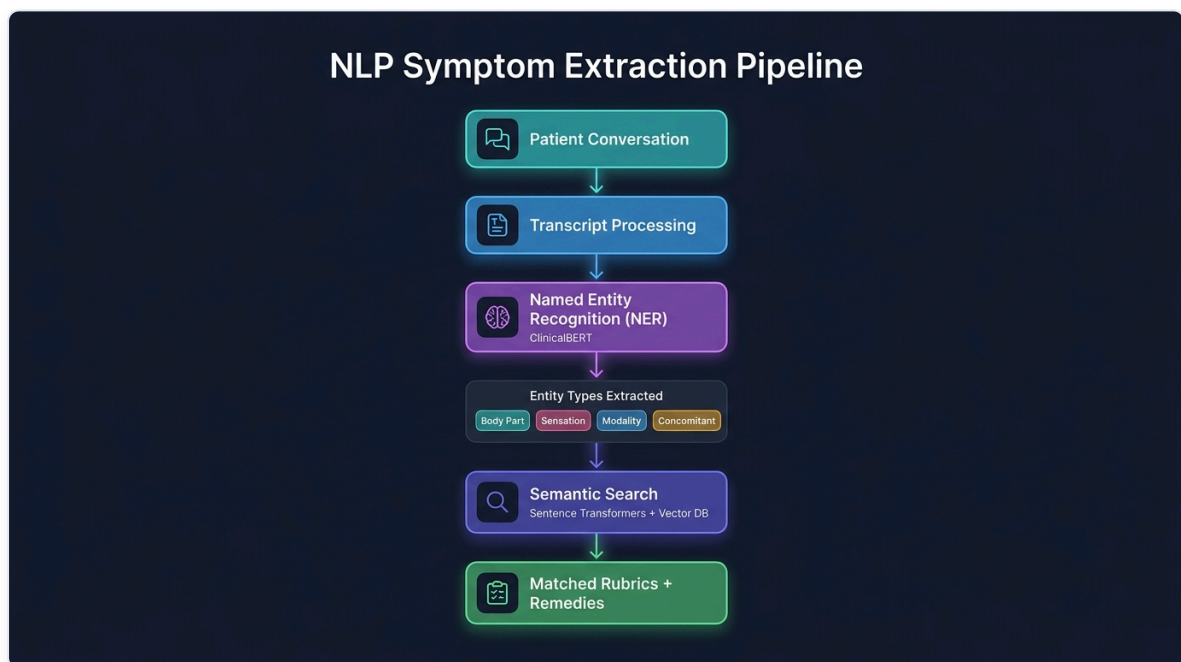


Figure: NLP Symptom Extraction Pipeline

Extraction Example

Patient says: "I have this terrible headache, mostly on the right side, it gets worse in the morning and feels like something pressing on my head."

Dimension	Extracted Value	Mapped Rubric Path
Chief complaint	Headache	HEAD > PAIN
Laterality	Right side	HEAD > PAIN > right side
Modality (time)	Morning	HEAD > PAIN > morning
Sensation	Pressing	HEAD > PAIN > pressing

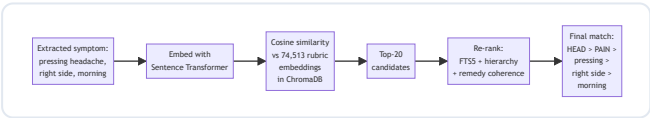
Model Selection: Why Transformers over RNNs

Criterion	RNN (LSTM/GRU)	BERT / Transformers
Context window	Degrades over long sequences	Full bidirectional attention
Pre-training	Must train from scratch	Transfer learning from medical corpora
Data efficiency	Needs 10K+ examples	Works with 2K–5K fine-tuning examples
Medical variants	None available	BioBERT, ClinicalBERT, PubMedBERT
NER performance	F1 ~78–85%	F1 ~88–95%
State of the art	No (2017 era)	Yes (2024–2026)

Decision: We use **ClinicalBERT** fine-tuned for homeopathic Named Entity Recognition, with **LLM-based extraction as a complementary rapid-prototyping tool**.

5.3 Semantic Rubric Matching

The core challenge: mapping extracted symptoms to the correct rubric among **74,513 candidates**.



Why semantic search over keyword search?

- Patient says "my head feels like it's in a vice" — no keyword match for "pressing"
- Semantic embeddings capture that "vice-like" is similar to "pressing" / "constricting"
- Hybrid approach: semantic similarity (recall) + keyword validation (precision)

5.4 Synthetic Case Generation

No public homeopathic case dataset exists with the granularity required for training. We generate our own using a **ground-truth-driven reverse pipeline**.



Figure: Synthetic Case Generation Strategy

Generation Pipeline

FOR each rubric combination sampled from the database:

1. Look up the remedy intersection from rubric_remedies
2. Prompt LLaMA 3 (8B):
 "Generate a realistic patient narrative for these rubrics: [list].
 Make it conversational and natural. Include modalities and sensations."
3. Store: (narrative, rubrics, remedies) as a training triple
4. Validate: Check that the narrative -> extraction -> matching pipeline
 recovers the original rubrics (self-validation)

Scale Targets

Phase	Cases	Method
Phase 1	5,000	LLM generation from rubric combinations
Phase 2	20,000	Augmentation (rephrase, register variation, noise injection)
Phase 3	500	Gold standard (validated against published case studies)

5.5 Doctor Dashboard

A **Streamlit web application** providing:

- **Patient report viewer** — Structured symptom summary with matched rubrics and remedies
- **Knowledge base search** — Full-text and semantic search across the entire repertory
- **Rubric tree explorer** — Navigate the 37-section hierarchy interactively
- **Remedy lookup** — Find all rubrics where a remedy appears, with grade information
- **Conversation transcript** — Review the full patient-chatbot interaction

6. Technical Stack

Component	Technology	Rationale
Knowledge Base	SQLite (existing) + ChromaDB	Structured queries + semantic vector search
Embeddings	all-MiniLM-L6-v2 / BioSentVec	Fast, accurate, medical-domain aware
Chatbot LLM	LLaMA 3 (8B) via Ollama	Local, private, zero API cost
Symptom NER	ClinicalBERT (HuggingFace)	Best medical NER performance
Synthetic data	LLaMA 3 + repertory DB	Ground-truth-driven, no external dependency
Backend	FastAPI (Python)	Clean API boundaries, async support
Frontend	Streamlit	Rapid prototyping, Python-native
Experiment tracking	MLflow / W&B	Reproducibility
Language	Python 3.10+	Unified ML + web stack

7. Implementation Roadmap

Phase 1: Foundation

- ☐ Generate embeddings for all 74,513 rubric paths
- ☐ Store in ChromaDB; build semantic search API
- ☐ Build synthetic case generator; generate initial 1,000 cases
- ☐ Define evaluation metrics: Rubric Match Accuracy, Remedy Recall@k, Conversation Completeness

Phase 2: Core ML

- ☐ Fine-tune ClinicalBERT on synthetic + hand-corrected data
- ☐ Build chatbot prototype with structured state tracking
- ☐ System prompt design for homeopathic case-taking
- ☐ End-to-end: conversation to extraction to rubric search
- ☐ Scale synthetic data to 5,000+ cases

Phase 3: Integration

- ☐ Build Streamlit doctor dashboard (report viewer, KB search, rubric explorer)
- ☐ End-to-end evaluation: simulate 50+ cases
- ☐ Benchmark time-to-diagnosis improvement

Phase 4: Research Output

- ☐ Write and submit paper(s)
- ☐ Open-source the dataset and tools
- ☐ Plan Materia Medica integration as future extension

8. Novel Research Contributions

This project offers **four distinct publishable contributions**:

8.1 First Dense Semantic Vector Index for Kent's Repertory

While [su4532](#) provides a raw digitized SQLite text database with keyword search, no existing open-access work has transformed this classical text into an **ML-ready semantic vector space**. We generate and open-source dense sentence embeddings across all 74,513 hierarchical rubric paths in ChromaDB, enabling neural semantic retrieval that bridges modern conversational patient complaints to 19th-century medical rubrics.

8.2 HomeoNER: Homeopathic NER Benchmark Dataset

No public annotated dataset exists for homeopathic symptom extraction. Our synthetic generation pipeline creates the **first benchmark dataset** for this task.

8.3 Conversational Case-Taking with Grounded Output

Existing chatbot research in homeopathy is surface-level (rule-based, no real evaluation). Our system combines LLM empathy with structured rubric grounding — a novel hybrid architecture.

8.4 Repertory Graph Analysis

The 1,923 cross-references in the database form a graph that has never been analyzed computationally. Network analysis can reveal latent symptom clusters and remedy affinities.

9. Literature Survey

Project / Paper	What they did	How we differ
HomeoGPT (Ghosh et al., 2024)	LLM-based homeopathic remedy suggestion and clinical dialogue	We ground reasoning in a verified 74k-rubric SQLite database rather than unconstrained generative outputs
HOHM Study (HOHM Foundation, 2025)	Evaluated AI remedy finders vs. live homeopaths on 100 acute cases (59% match rate)	We focus on pre-consultation case-taking and rubric mapping assistance, not autonomous prescribing
AI-Aided Homeopathic Clinic (JDDT, 2026)	Conceptual multi-modal framework (voice tone, facial expression, language)	We focus first on robust text-based NLP and semantic rubric retrieval; multi-modal is future work
Symptom-BERT (Zeinali et al., 2024)	Pre-trained and fine-tuned BERT for symptom NER in EHR clinical notes	We adapt and fine-tune for homeopathic vocabulary (modalities, sensations, concomitants, locations)
Synthea (GitHub) (Walonoski et al., 2018)	Synthetic patient record generator for conventional clinical scenarios	We build a domain-specific synthetic case generator grounded in combinatorial repertory rubrics
BioBERT / ClinicalBERT (2019–2020)	Domain-adapted transformers for biomedical and clinical text	We fine-tune these architectures on our HomeoNER corpus for homeopathic symptom NER

10. Risk Assessment

Risk	Impact	Mitigation
OCR errors in rubric labels	Incorrect rubric matches	DB has 7,510 flagged issues; prioritize high-traffic rubrics
Unverified remedy grades	<code>grade</code> is NULL; only <code>grade_candidate</code> exists	Use as soft signal; document limitation
LLM hallucination	Fabricated rubrics or remedies	Always validate against DB; never trust ungrounded output
No real patient data	Cannot validate against live cases initially	Synthetic data + published case studies as benchmarks
LLaMA 8B capacity limits	May struggle with complex extraction	Use structured prompting; complement with fine-tuned ClinicalBERT

11. References

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